F172 M aircraft involved in an accident, and the application has generated a detailed, context-rich response.

With the RAG system in place, users can explore the indexed corpus, obtain detailed information, and gain a deeper understanding of the topics covered. This lays the foundation for the last piece of the web application, which is the introduction of grounding support.

7.2.6 Adding grounding support

Grounding is a valuable technique in generative AI that ensures AI-generated responses are validated by reliable, factual information sources. Essentially, grounding provides another layer of verification. Traditional AI without grounding resembles a student who relies entirely on their own memory to answer a question, potentially mixing up or fabricating information with no way to verify its accuracy. With grounding, you have something more like a very smart librarian who doesn't just answer questions from memory but always checks multiple reference books before giving an answer.

Technically, the grounding process can be visualized as a sequential flow: the user poses a question, which triggers a search through relevant documents. The system extracts precise information, generates a response that incorporates these verified sources, annotates the answer with its origins, and then delivers a verified response. Figure 7.10 shows the pieces of this process.

GROUNDING TYPES

There are five main types of grounding. Each method serves a unique purpose in integrating external knowledge while maintaining consistency:

- *Retrieval-augmented grounding*—Retrieves relevant information from external sources and integrates it into the generation process, which reduces the chance of inaccuracies. As an example, Google recently added grounding support on a limited basis using Google search or via your own data.
- *Knowledge base grounding*—Uses curated, validated information repositories to constrain responses to specific, trusted domains.
- *Citation-based grounding*—Generates responses with explicit source references, allowing users to verify the credibility of the information. For example, in academic writing or research-focused AI applications, when an AI generates a summary or analysis, it may include citations from peer-reviewed journals or other databases.
- *Contextual grounding*—Adapts responses based on previous interactions and modifies the output to the user's specific needs. For instance, if a user previously asked about a specific product, the chatbot can reference that interaction in future conversations.
- *Log probability grounding*—Uses log probabilities to assess the confidence of a language model's responses. By calculating the probability of each token